Here typically we have α motoneuron that gets information from the spinal cord to the ventral

roots, to the muscles. Here we have a very simple stretching reflex loop, because information at the level of the spinal cord is not just afferent to muscles, but the muscles spindle is sending projections and information to the spinal cord about the position of the muscle and the posterior sensitive elements is connected to the anterior one, through the inhibitory interneuron that connects the two neurons. So, the ventral nerve roots convey efferent motor signals, and the dorsal roots are getting the afferent sensitive signal.Typically, the muscles spindles (sensory receptors that detect the amount and weight of stretch
in muscles) transmit information to the distal processing of sensory neurons and convey to the spinal grey matter through the dorsal root. Here we have the inhibitory interneuron that connects to the anterior motoneuron, and this project to the muscle. So, this local circuit controls what you typically experience, if you want to extend your arm you have to use one group of muscle, but you
should inhibit the antagonist group of muscle that is contrasting the action. When the neurologist hit the tendon on the knee, it stretches the spindle and you are performing a stretch and providing
information that that part of the muscle is stretching, and this is going to induce an opposite reaction (the extension of the leg). So, what you are actually testing is this type of circuit, that has
to inhibit or excite agonist and antagonist muscles of our limbs. These kind of stretching reflexes,
are also useful when the stimulus is not starting from the spindle but is starting from the skin, just as with pain, where you have a painful stimulation, and before the information gets to the cortex you just remove the limb from the possible source of damage.
The control effect that the cortical motoneurons are performing to the spinal cord is very important, because is regulating the excitatory inhibitory balance. Most of the time, when you have a lesion at the level of the spinal cord, and this is no more able to get information from the brain, muscles are going in a sort of like conflicting presence of stimuli of excitatory and inhibition, and without this element of control you see that the muscle is primarily stretching and detecting the position that we call plasticity, is excessing excitatory response.
This kind of complete pathway from the cortex down to the spinal cord is fundamental to maintain
the proper functions of the muscles. --- explain

The stretch (myotatic) reflex is a fast spinal **negative-feedback** loop that stabilizes muscle length and contributes to normal muscle tone. It works because sensory signals from muscle spindles can excite the motor neurons of the stretched muscle while, via interneurons, inhibiting the antagonist muscle.[1] [2]

Stretch reflex wiring

When a muscle is rapidly stretched (e.g., tendon tap), intrafusal fibers in the muscle spindle are stretched and increase firing in large Ia afferents. Those Ia afferents make a direct (monosynaptic) excitatory connection onto α-motoneurons in the ventral horn that innervate the same (homonymous) muscle, causing it to contract and resist the stretch. In parallel, Ia afferents recruit local inhibitory circuitry so α-motoneurons of antagonist muscles are inhibited (reciprocal inhibition), allowing smooth joint motion instead of co-contraction.[2] [3]

Agonist vs antagonist control

This is why, during a voluntary action like elbow extension, you can increase drive to the triceps while "turning down" the biceps at the spinal level: reciprocal inhibition prevents the antagonist from counteracting the intended movement. Clinically, the patellar tendon reflex is largely a test of this short-latency spindle → Ia afferent → α-motoneuron pathway and its coordinated inhibition of antagonists.[2]

Skin/pain withdrawal reflex

Not all protective reflexes start from spindles: painful cutaneous input can enter the spinal cord and, through interneurons, excite flexor α-motoneurons while inhibiting extensor α-motoneurons to withdraw the limb quickly (a polysynaptic withdrawal reflex). This happens before cortical processing, which is why you can pull away from a harmful stimulus so fast.[4]

Descending (cortical) modulation and lesions

Descending pathways from the brain modulate spinal motor pools and reflex "gain," effectively setting the desired muscle length/level of tone that the stretch reflex will defend. The γ-motoneuron system also helps by keeping spindles taut during voluntary contraction (α–γ coactivation), so spindles keep providing usable feedback across muscle lengths. When descending control is disrupted (e.g., upper motor neuron/spinal pathway lesions), reflexes and tone can become abnormally increased (hyperreflexia/spasticity) because spinal circuits are less regulated, which is exactly why tendon reflexes are informative in neurological exams.[5] [2]

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